IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL AND DATA-DRIVEN CONTROL SYSTEMS

Tuesday, 12 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Y. Gao
Second Marker(s) : D. Angeli

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ELEC70090: Digital and Data-Driven Control Systems

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Information for candidates:

- $Z(\frac{1}{s}) = Z(k1(kT)) = \frac{z}{z-1}$
- $Z(\frac{1}{s^2}) = Z(k1(kT)) = \frac{Tz}{(z-1)^2}$
- $Z(\frac{1}{s+a}) = Z(e^{-akT}) = \frac{z}{z-e^{-aT}}$
- Transfer function of the zero-order hold: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Definition of w-plane: $z = \frac{1+\frac{wT}{2}}{1-\frac{wT}{2}}$, $w = \frac{2}{T}\frac{z-1}{z+1}$
- Note that, for a given signal r (also indicated as $r(k)$, $r(kT)$ or $r(t)$), $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.
- Consider the characteristic equation is $1 + KF(z) = 1 + K\frac{N(z)}{D(z)}$. Suppose that $N(z)$ has roots z_i, $i = 1, \ldots, m$, $D(z)$ has roots p_i, $i = 1, \ldots, n$. The difference between n and m is the relative degree $r = n - m$.

  If $r > 0$ there are asymptotes of the root locus that intersect the real axis at $\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{r}$, and radiate out with angles $\theta = \pm q\frac{\pi}{r}$, where $q = 1, 3, 5, \ldots$.

  Break-away or break-in points of the root locus exist where $N(z)D'(z) - N'(z)D(z) = 0$ ($'$ indicates the derivative).

1(A). Show that

$$Z\left[\sum_{h=0}^{k} x(h)\right] = \frac{1}{1-z^{-1}}X(z),$$
$$Z\left[\sum_{h=0}^{k-1} x(h)\right] = \frac{z^{-1}}{1-z^{-1}}X(z).$$

[4 marks]

1(B). Compute the inverse Z-transform of

$$X(z) = \frac{z(z+2)}{(z-1)^2}.$$

[4 marks]

1(C). Consider the rectangular region in the s-plane defined by

$$-\sigma_2 \leq Re(s) \leq -\sigma_1 \text{ and } -\omega_2 \leq Im(s) \leq \omega_1,$$

with $\sigma_2 > \sigma_1 > 0$, $\omega_s/2 > \omega_2 > 0$, and $\omega_s/2 > \omega_1 > 0$ with $\omega_2 \neq \omega_1$. Determine the image of this region in the z-plane resulting from the mapping $z = e^{sT}$. Here $\omega_s = \frac{2\pi}{T}$. [4 marks]

1(D). Consider a position control system where the plant is a DC motor modeled by the transfer function:

$$P(s) = \frac{1}{s(s+1)}$$

(i) Suppose that the continuous-time system is controlled a proportional controller K. Determine for which values of K the closed-loop continuous-time system is asymptotically stable. [2 marks]

Now suppose that the system is controlled by a digital proportional controller K through a Zero-Order Hold (ZOH) with a sampling time $T_s = 1$ second.

(ii) Compute the equivalent discrete-time model $HP(z)$ for the plant with the hold and sampler. [3 marks]

(iii) Determine the range of K for which the closed-loop discrete-time system is stable. [4 marks]

(iv) Sketch the root locus of the closed-loop system for $K > 0$. Your sketch must clearly indicate:

- The open-loop poles and zeros.
- The segments of the real axis that belong to the root locus.
- The breakaway point (calculate its approximate location).

[4 marks]

2(A). The continuous-time system is defined by

$$\begin{aligned} \dot{x}(t) &= Fx(t) + Gu(t), \\ y(t) &= Lx(t), \end{aligned} \tag{1}$$

with

$$F = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad L = \begin{bmatrix} 2 & 0 \end{bmatrix}.$$

Let the sampling time be T seconds.

(i) Discretize the continuous-time state-space equations (1). To this end, determine the matrices A, B, and C of the representation

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k), \\ y(k) &= Cx(k), \end{aligned} \tag{2}$$

when the input $u(t)$ is sampled and fed to a zero-order hold. [4 marks]
Hint: recall that $e^{Ft} = \mathcal{L}^{-1}[(sI-F)^{-1}]$*, where* $\mathcal{L}^{-1}$ *indicates the inverse Laplace transform operator.*

(ii) Compute the input-output transfer function of the discrete-time system (2) from the matrices A, B, and C. [4 marks]

(iii) The continuous-time state-space system (1) has the input-output transfer function

$$\frac{Y(s)}{U(s)} = P(s) = L(sI - F)^{-1}G = \frac{2}{s(s+2)}.$$

Compute the equivalent discrete-time model $HP(z)$ for the plant $P(s)$ interconnected to the zero-order hold and the sampler. Show that $HP(z)$ is the same transfer function as the one obtained in part (ii). [4 marks]

(iv) Calculate the controllability matrix $\mathcal{C}$ for the continuous-time system (F, G) and determine its rank. Based on this result, explain whether the equivalent discrete-time system (A, B) is controllable for all sampling times $T > 0$. [4 marks]

(v) Calculate the observability matrix $\mathcal{O}$ for the continuous-time system using the output matrix $L = \begin{bmatrix} 2 & 0 \end{bmatrix}$. Determine if the system is observable. Does the discrete-time system (A, C) retain this property? [4 marks]

2(B). Consider the discrete-time system (2) with

$$A = \begin{bmatrix} 1 & 0.4 \\ 0 & 0.6 \end{bmatrix}, \quad B = \begin{bmatrix} 0.1 \\ 0.4 \end{bmatrix}$$

Design a state feedback control law $u[k] = -Kx[k]$ with gain matrix $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$ to place the closed-loop poles at the complex locations $z_{1,2} = 0.5 \pm j0.2$.

[5 marks]

3(A). Consider a scalar linear discrete-time system given by the dynamics:

$$x(k+1) = ax(k) + bu(k) + w(k)$$

where $x(k) \in \mathbb{R}$ is the state, $u(k) \in \mathbb{R}$ is the control input, and $w(k)$ is an independent and identically distributed (i.i.d.) Gaussian noise sequence with $w(k) \sim \mathcal{N}(0, \sigma^2)$. Here, $a, b \neq 0$.

Give a finite horizon N, the objective is to minimize the finite-horizon cost function:

$$J = \mathbb{E}\left[sx^2(N) + \sum_{k=0}^{N-1} (qx^2(k) + ru^2(k)) \right]$$

where $s, q \geq 0$ and $r > 0$.

(i) Define the cost-to-go function (value function) $V_k(x(k))$ for the time step k. [2 marks]

(ii) Derive, using the dynamic programming, the optimal control $u(k)$ that minimizes $J(x(0))$ and determine the closed-form expression for the minimal total cost $J_{\min}(x(0))$ (i.e., $V_0(x(0))$). [10 marks]

3(B). Consider the discrete–time LTI system

$$x(k+1) = Ax(k) + Bu(k), \qquad y(k) = Cx(k) + Du(k), \tag{3}$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^m$, and $y(k) \in \mathbb{R}^p$. Assume that the pair (A, B) is controllable.

Let $\{u_d(k), y_d(k)\}_{k=0}^{T-1}$ be a measured input–output trajectory generated by the system (3). For a signal $z(k) \in \mathbb{R}^q$, define the block Hankel matrix

$$Z_{0,t,T-t+1} = \begin{bmatrix} z(0) & z(1) & \cdots & z(T-t) \\ z(1) & z(2) & \cdots & z(T-t+1) \\ \vdots & \vdots & \ddots & \vdots \\ z(t-1) & z(t) & \cdots & z(T-1) \end{bmatrix}.$$

(i) Define what it means for the input $u_d[0, T-1]$ to be *persistently exciting of order t*. [3 marks]

(ii) Assume that $u_d[0, T-1]$ is persistently exciting of order $n+t$. Show that for every length–t input–output trajectory $\{u(k), y(k)\}_{k=0}^{t-1}$ of system (3), there exists a vector $g \in \mathbb{R}^{T-t+1}$ such that

$$\begin{bmatrix} U_{0,t,T-t+1} \\ Y_{0,t,T-t+1} \end{bmatrix} g = \begin{bmatrix} u(0) \\ \vdots \\ u(t-1) \\ y(0) \\ \vdots \\ y(t-1) \end{bmatrix}.$$

Hint: If (A, B) is controllable and the input $u_{d,[0,T-1]}$ is persistently exciting of order $n+t$, then the stacked matrix has full row rank:

$$\mathit{rank} \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix} = n + tm.$$

[5 marks]

(iii) Let $X_{0,T} := [\, x_d(0) \ \cdots \ x_d(T-1) \,]$ and $X_{1,T} := [\, x_d(1) \ \cdots \ x_d(T) \,]$. Assume that

$$\operatorname{rank} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix} = m + n.$$

Show that a data-driven representation of the open-loop system dynamics is

$$x(k+1) = X_{1,T} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix}^{\dagger} \begin{bmatrix} u(k) \\ x(k) \end{bmatrix}$$

where $[\cdot]^{\dagger}$ is the Moore-Penrose inverse. [5 marks]